

Introduction

Objective:

This project analyzes the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using Hyperliquid historical trading data. The goal is to identify patterns in profitability, win rates, and trading behavior under different market sentiment conditions.

Datasets Used

Dataset 1: Bitcoin Fear & Greed Index

- Date
- Value
- Classification (Fear, Greed, Extreme Fear, Extreme Greed, Neutral)

Dataset 2: Hyperliquid Historical Trading Data

- Coin
- Side
- Closed PnL
- Fee
- Timestamp
- Trade Information

Data Preparation

The datasets were cleaned and merged using the date field. Timestamp data was converted into a standard date format, and missing sentiment values were removed before analysis.

Visualizations

Figure 1: Profit by Sentiment

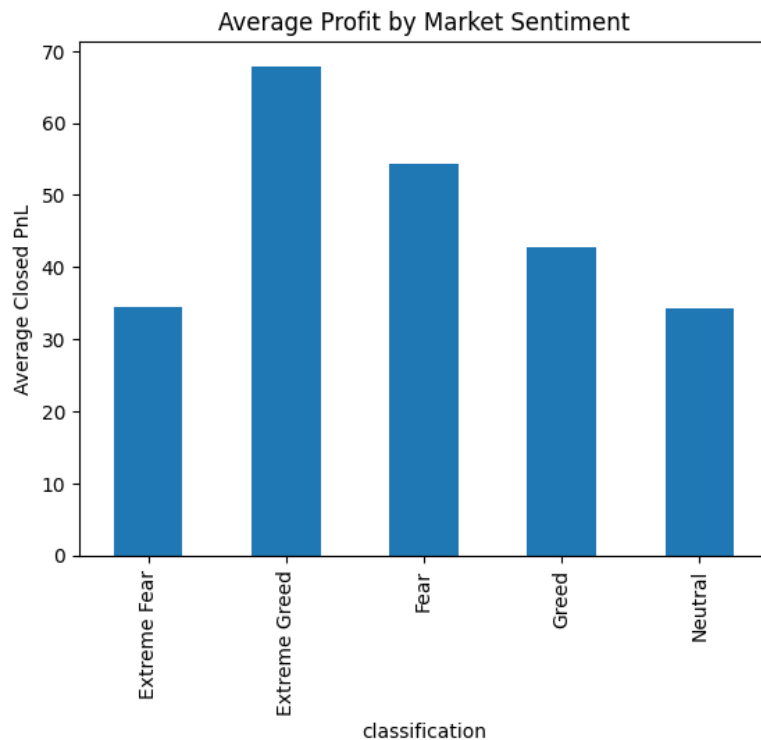


Table 1: Average Profit by Market Sentiment

Sentiment	Average Closed PnL
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

Explanation:

The chart shows the average profit (Closed PnL) achieved under different market sentiment conditions.

Figure 2: Win Rate by Sentiment

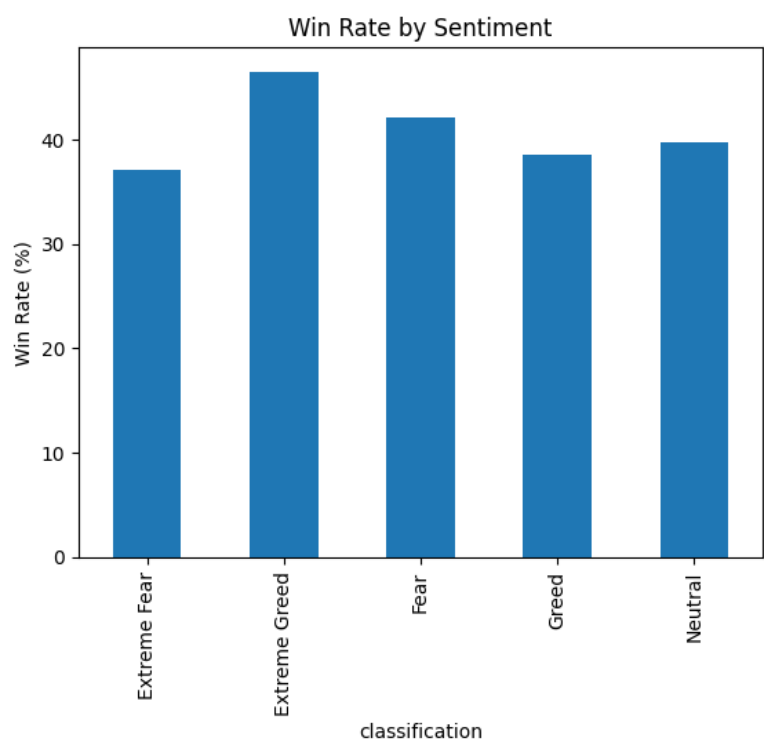


Table 2: Win Rate by Market Sentiment

Market Sentiment Win Rate (%)

Extreme Greed	46.49
Fear	42.08
Neutral	39.70
Greed	38.48
Extreme Fear	37.06

Explanation:

The chart illustrates the percentage of profitable trades across sentiment categories.

Figure 3: Trading Volume by Sentiment

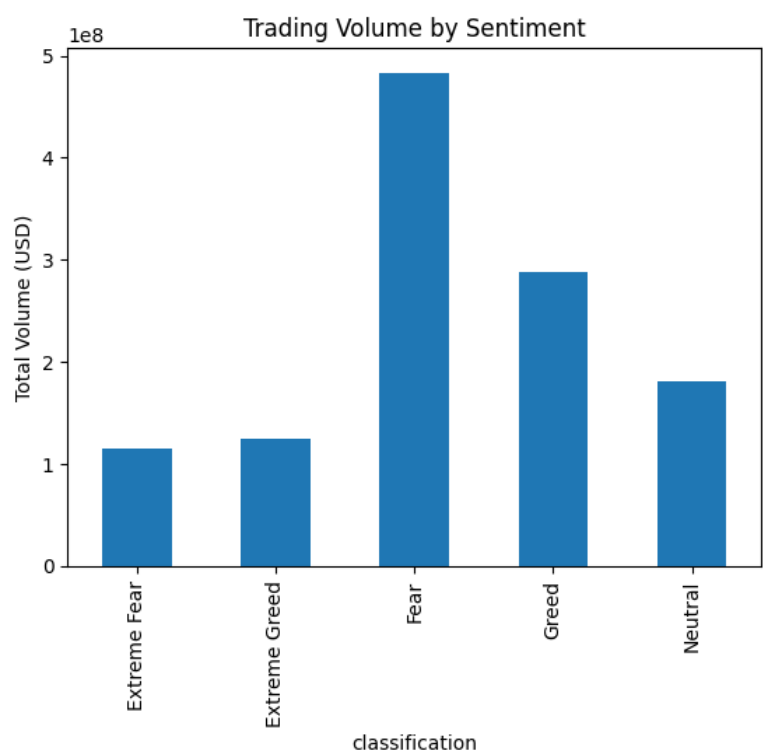


Table 3: Trading Volume by Market Sentiment

Market Sentiment Total Trading Volume (USD)

Fear	483,324,800
Greed	288,582,500
Neutral	180,242,100
Extreme Greed	124,465,200
Extreme Fear	114,484,300

Explanation:

The chart displays total trading volume across different sentiment conditions...

Figure 4: Distribution of Closed PnL

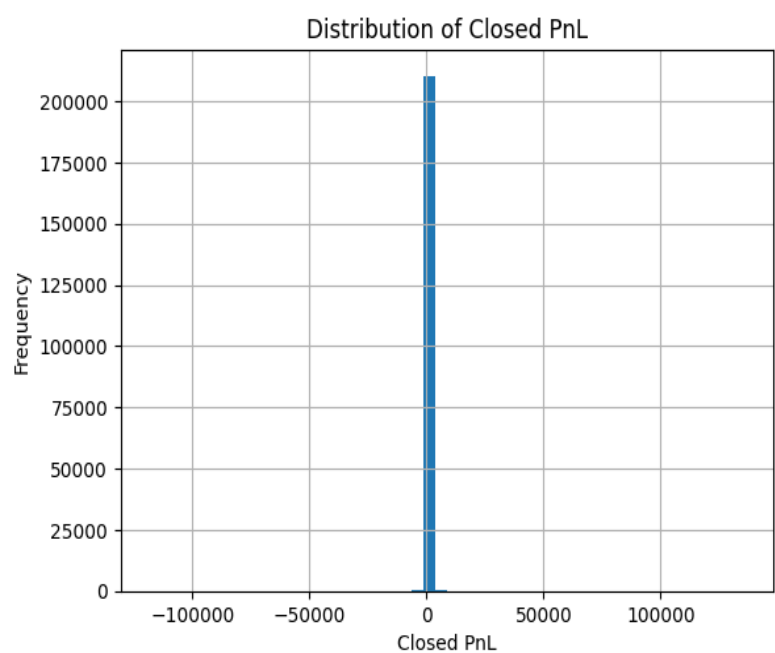


Table 4: Distribution Analysis

Observation	Description
Distribution Shape	Most trades are concentrated around smaller profit/loss values.
Outliers	A smaller number of trades generated very large profits or losses.
Trading Pattern	Profitability varies significantly across trades, indicating varying risk-return profiles.

Explanation:

The histogram shows the distribution of trade profits and losses.

Key Findings

1. **Extreme Greed** market conditions produced the highest average profit, with an average Closed PnL of **67.89**.
2. **Fear** conditions generated the second-highest average profit, with an average Closed PnL of **54.29**.
3. Traders achieved the highest win rate during **Extreme Greed** periods (**46.49%**), indicating stronger trading performance during highly optimistic market conditions.
4. **Extreme Fear** periods recorded the lowest win rate (**37.06%**), suggesting reduced trading success during highly pessimistic market conditions.
5. Analysis of trading activity showed that **SELL trades exceeded BUY trades during Greed and Extreme Greed periods**, indicating potential profit-taking behavior when market sentiment is positive.

Conclusion

The analysis indicates that market sentiment has a measurable impact on trader performance. Traders achieved higher profitability and win rates during optimistic market conditions, particularly Extreme Greed periods. These findings suggest that sentiment indicators can provide valuable context when evaluating trading behavior and performance.